# Quiz 2: Practice Solutions

Lectures 4–7

Try the practice handout in 15 minutes before reading these solutions. Each of its five parts earns 2 practice points for a correct final answer and 0 otherwise.

| 1(a) | 1(b) | 1(c) | 2(a) | 2(b) |
|---|---|---|---|---|
| 13 | $-4$ | 3 | C | D |

## 1. Value iteration, evaluation, and convergence — 6 points

### (a) One value-iteration update

Use $V_k = (4, 8)$, average successors, then maximize over actions:

$$V_{k+1}(\mathrm{Y}) = \max\left\{10 + \tfrac{1}{2}\left(\tfrac{1}{2}\cdot 4 + \tfrac{1}{2}\cdot 8\right),\ -1 + \tfrac{1}{2}\cdot 4\right\} = \max\{13, 1\} = \boxed{13}.$$

### (b) Exact evaluation of the fixed policy $(b, b)$

Write $x = V^\pi(\mathrm{X})$ and $y = V^\pi(\mathrm{Y})$. Following $b$ forever gives

$$x = -4 + \tfrac{1}{2}y, \qquad y = -1 + \tfrac{1}{2}x \quad\Longrightarrow\quad y = -3 + \tfrac{1}{4}y \quad\Longrightarrow\quad \boxed{y = -4} \quad (x = -6).$$

There is no maximization in policy evaluation. A single update from $(4, 8)$ would not evaluate this policy exactly.

### (c) Two rounds of the PI rate

Apply the guarantee once per greedy improvement:

$$E_2 \le \gamma E_1 \le \gamma^2 E_0 \le \left(\tfrac{1}{2}\right)^2 12 = \boxed{3}.$$

This is an upper bound on the new policy's loss, not its exact loss. An exact evaluation computes the current policy's values; it does not by itself make the policy optimal.

## 2. Policy improvement and stopping — 4 points

### (a) C: Choose $b$ at X and $a$ at Y

Greedy improvement chooses the action with the larger action value at each state:

$$\mathrm{X}: \quad 3 > 2 \ \Rightarrow\ \pi_1(\mathrm{X}) = b, \qquad \mathrm{Y}: \quad 14 > 0 \ \Rightarrow\ \pi_1(\mathrm{Y}) = a.$$

Thus $\pi_1 = (b, a)$, choice **C**. The action values are already given; no Bellman backup or policy evaluation is needed.

### (b) D: Both I and II

**I:** the equality $\mathcal{T}V^\pi = V^\pi$ makes $V^\pi$ a fixed point of the Bellman optimality operator. Its unique fixed point is $V^\star$, so $V^\pi = V^\star$ and $\pi$ is optimal.

**II:** VI uses $V_{k+1} = \mathcal{T}V_k$, so equality of the two tables gives $V_k = \mathcal{T}V_k = V^\star$. A policy greedy with respect to $V^\star$ is optimal.

The equality in II is about *values*. The same greedy actions can be selected from two different, nonoptimal tables. Thus unchanged greedy actions alone are not a valid VI stopping certificate.